

8 (**short**) A faulty bit in a computer memory changes state in any given clock cycle with probability p .

(a) Find the probability that it will be in the same state two clock cycles later. [5]

(b) If P_N is the probability that it is in the same state N cycles later, show that

$$P_{N+1} = p + (1-2p)P_N \quad [5]$$

9 (**short**) The function $f(x, y)$ is given by

$$f(x, y) = x^2 + 4xy + 2y^2 - 6x - 8y$$

Find and classify the stationary points of f . [10]